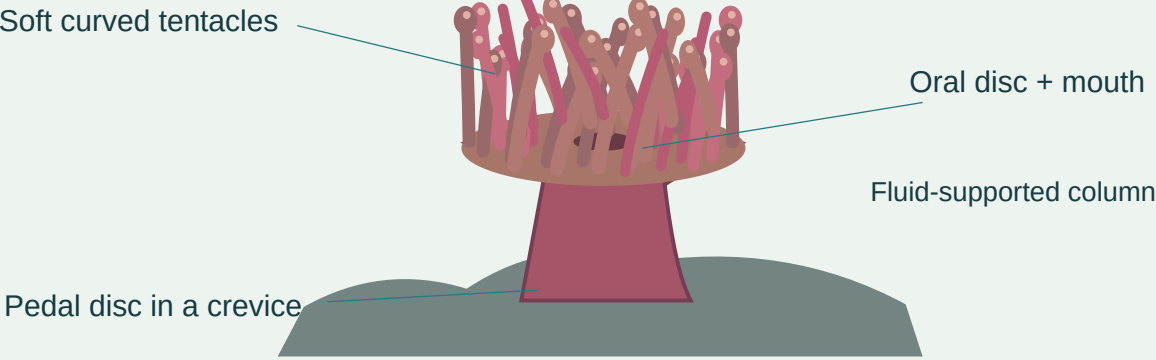


Sea anemones

Slow journeys. Flexible tentacles. A fluid-supported body.

BUBBLE-TIP · *Entacmaea quadricolor*



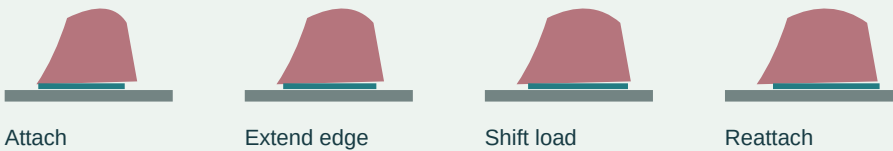
Soft tentacles, not spines

Muscular tissue bends, shortens and extends against fluid pressure. Stinging cells are microscopic; the visible crown is soft. [6]

Our focus: *Entacmaea quadricolor*, the bubble-tip. Dense, uneven tentacles can mix swollen and slender tips. The foot is usually hidden in a rock crevice. [1]

BIOLOGY FIRST · SCHEMATICS BELOW

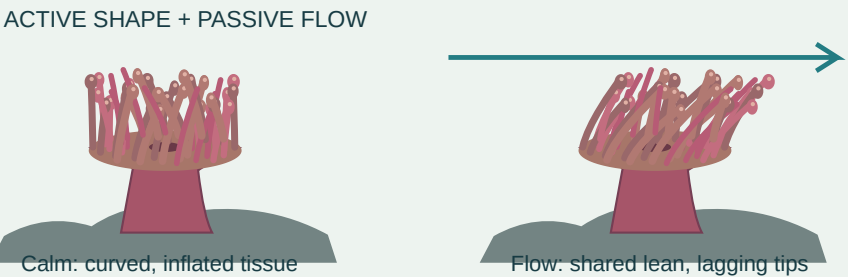
01 · A foot that can relocate



Adults can move to new positions in an aquarium. Daily observations confirm relocation, but do not give a crawling speed. [2]

Foot adhesion is studied in *Exaiptasia*. [4] This moving-attachment sequence is a visual approximation, pending bubble-tip footage.

02 · Flow gives a shared rhythm



Field observations of *Aiptasia diaphana* show downstream bending and wave-linked flutter; long tentacles make larger excursions. [3]

For sheltered bubble-tips: gentle coherent lean, curved tips and local lag. This transfer is a modeling inference.

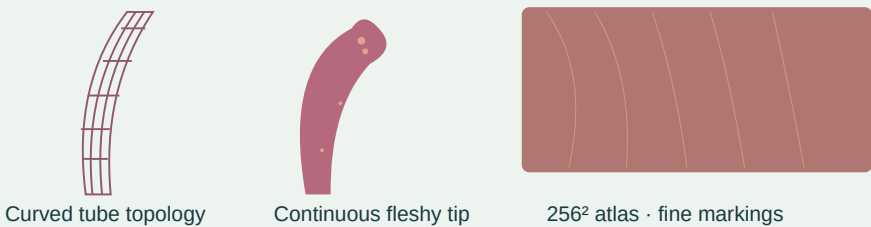
03 · Muscles change the shape



Layer active shortening and recovery onto water motion. Roots remain attached to the disc; tips respond more visibly. [6]

Stomphia can detach and swim after particular predator stimuli. [5] That is a separate species' behavior, not the bubble-tip default.

04 · A continuous fleshy surface



Use curved tube geometry, blunt tips, uneven bulbs and a dense crown. Striation, speckles and translucency are documented in bubble-tips. [1]

256² texture atlas is a proposed game choice. Fine markings belong in the texture; silhouette and bending need geometry.

05 · Small Forth ideas, richer motion

```
\ Normalized root-to-tip coordinate
: tip-weight ( u -- w ) dup * ;
\ First-order response; tau > 0
: lag-step ( x target dt tau -- x' )
  / 1 min 0 max >r over -
  r> * + ;
```

One shared water signal; length-dependent tip response; sparse local contractions. Avoid one actor or mailbox loop per tentacle.

Proposed controls: D-pad crawls on valid rock; hold A withdraws, release allows recovery. Switch control focus in settings. No fish-like dart.

These fragments are illustrative, not a complete rig. Speeds, periods and response times need footage and profiling.

Research saved first: [sea-anemone-research.md](#) · Checked 5 October 2026.

SOURCES

[1] Titus et al. · bubble-tip morphology (2024) · [2] Watson et al. · bubble-tip relocation (2022) · [3] Full et al. · Aiptasia field flow, SICB abstract (2013) · [4] Clarke et al. · Exaiptasia adhesion (2020) · [5] Robson · Stomphia swimming (1961) · [6] Cnidarian muscles review (2017)

A3 portrait · 297 × 420 mm · Original editable vector schematics, not photographs or implemented assets. Cross-species evidence is labelled. Biology and proposed game behavior are distinct.